

no matter when an investor entered the fund. In ETFs, your tax basis depends only on when you bought the ETF.

3. More transparency

ETF holdings are published daily. Mutual funds report their holdings quarterly. The daily disclosure is actually an impediment for *actively managed ETFs*, which do not blindly track indices; why would you want to allow everyone to see what you're doing if you have a secret sauce?

4. Can be shorted

Mutual funds are long-only products. Because ETFs trade on exchanges, they can be shorted. Given the underperformance of active mutual funds, shorting actively managed ETFs might be a way to capitalize on the active management discount.

I see two major disadvantages of ETFs for individuals. First, the fact that you can easily trade means investors trade often. Odean (1999) and Barber and Odean (2000) show that individuals lose money when they trade a lot because they trade pro-cyclically: they are reluctant to sell losing stocks and are too quick to realize their winners. Second, the large assortment of ETFs can lead to some investors becoming attracted to overly narrow products that do not give them adequate diversification (see chapter 3).

5.2. FAIR PRICING

ETF prices quoted on an exchange should correspond closely with the NAV through *no-arbitrage pricing*. ETF shares are created when an *authorized participant* (brokers or institutional investors, which must be registered with the ETF) deposits a *creation basket* with the ETF. The creation basket is a portfolio of the underlying securities in the ETF, usually reflecting the index the ETF is tracking. (In some cases, small amounts of cash can be substituted for securities that are hard to obtain.) In return for depositing securities, the authorized participant receives a large block of shares, like 25,000 or even 100,000, which is the *creation unit*. The authorized participant can then break up creation units and sell the smaller blocks on the exchange. The transactions between the authorized participant and the fund are deemed *in-kind transactions* and are not subject to tax.

To do the reverse process, a creation unit can be disposed of by selling back the shares to the ETF. In return, the authorized participant receives the creation basket.

If the ETF is trading below its NAV during the day, authorized participants can buy low and sell high. They buy shares of the ETF on the market and then offer them to the ETF and receive the creation basket. As they buy ETF shares, the price of the ETF rises until the ETF price reflects the NAV. Conversely if the ETF trades above its NAV, authorized participants sell the ETF (which is expensive) and buy